The Productivity of Employees and Contractors: Evidence from the Emergency Department

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Employees (85%) vs Contractors (15%)

- **Legal Agreements**: e.g., taxes, protections, benefits, liability...
- **Conditions**: e.g., schedules, location...
- **Culture**: e.g., insiders vs outsiders
- **Selection**: i.e., ability, preferences...

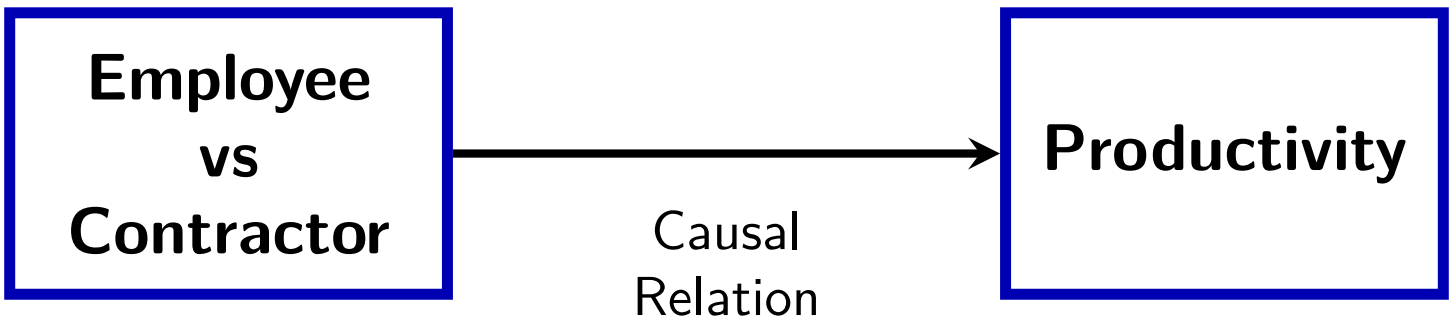
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Work on Contractors

- **Measurement:** e.g., Katz and Krueger (2019), Abraham et al. (2023)
- **Preferences for Alternative Work Arrangements:** e.g., Mas and Pallais (2020)
- **Organising the Gig Economy:** e.g., Stanton and Thomas (2016)

This paper:

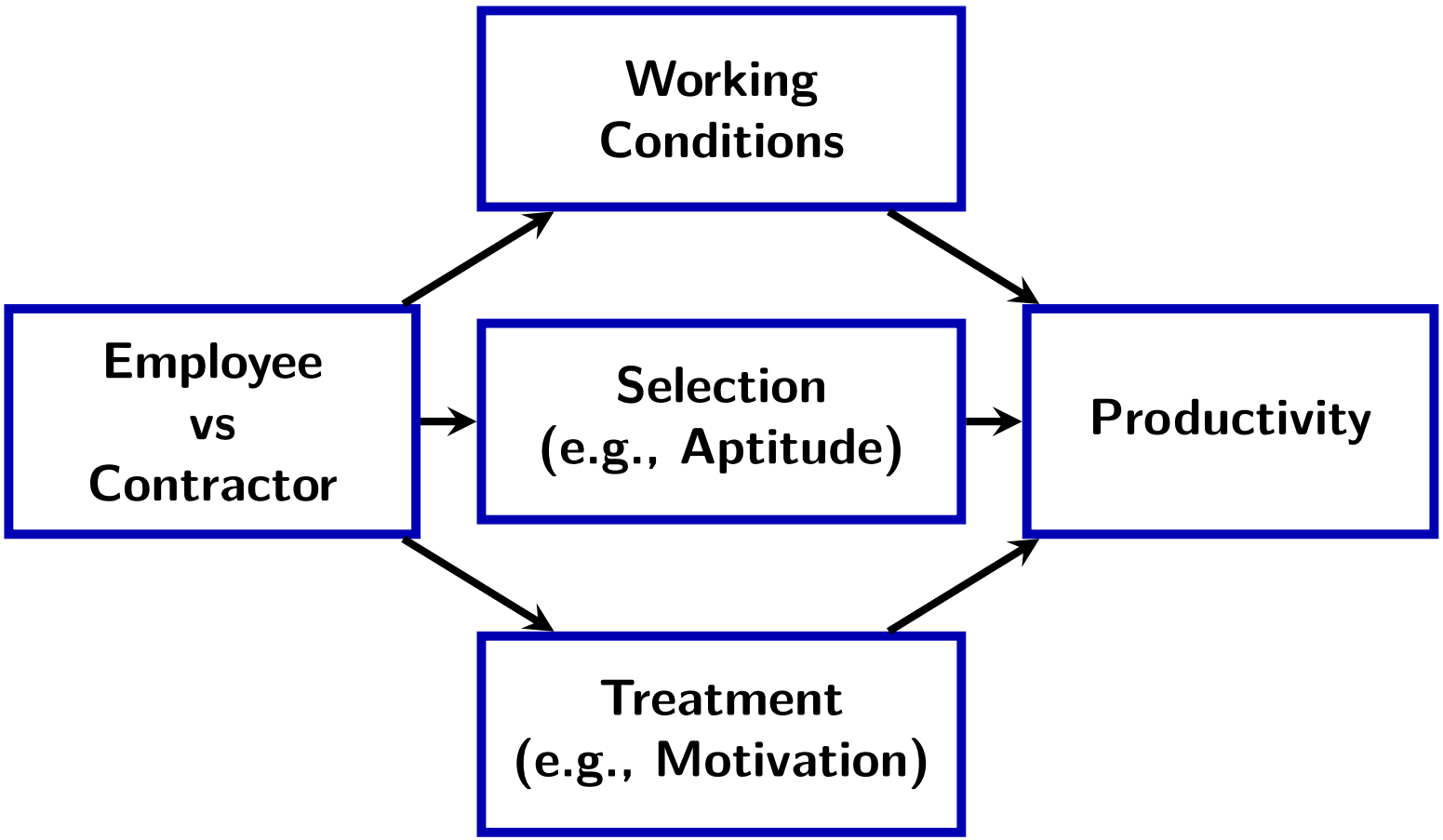


Little progress on this question:
hard to observe contractors and employees doing the same tasks

⇓

Setting: doctors in an emergency department

Worker Type is a Bundle of Characteristics:



Road Map:

Working Conditions → Setting holds **tasks and conditions** fixed, patients **exogenously assigned**
$\Rightarrow$ compare the two types on the same work

Selection

Treatment

Road Map:

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Selection → Policy: contractors transition into employees
⇒ exploit within-doctor variation

Treatment

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Working Conditions → Setting holds **tasks and conditions** fixed, patients **exogenously assigned**
⇒ compare the two types on the same work

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Treatment → Contract status itself changes behaviour
⇒ what is the mechanism?

Preview of Results: Employees Perform Better

~~**Working Conditions**~~

~~**Selection**~~

Treatment → **Mechanism:** increased willingness to cooperate with or help **other employees**

Setting

Niguarda Hospital, Milan

- large and good (37th in Newsweek 2025)

Doctors in Emergency Department

- individual high-skill high-stakes work
- organised into pods (i.e., sets of beds)
- rigid working shifts
- exogenous assignment of patients to doctors

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Dipendente

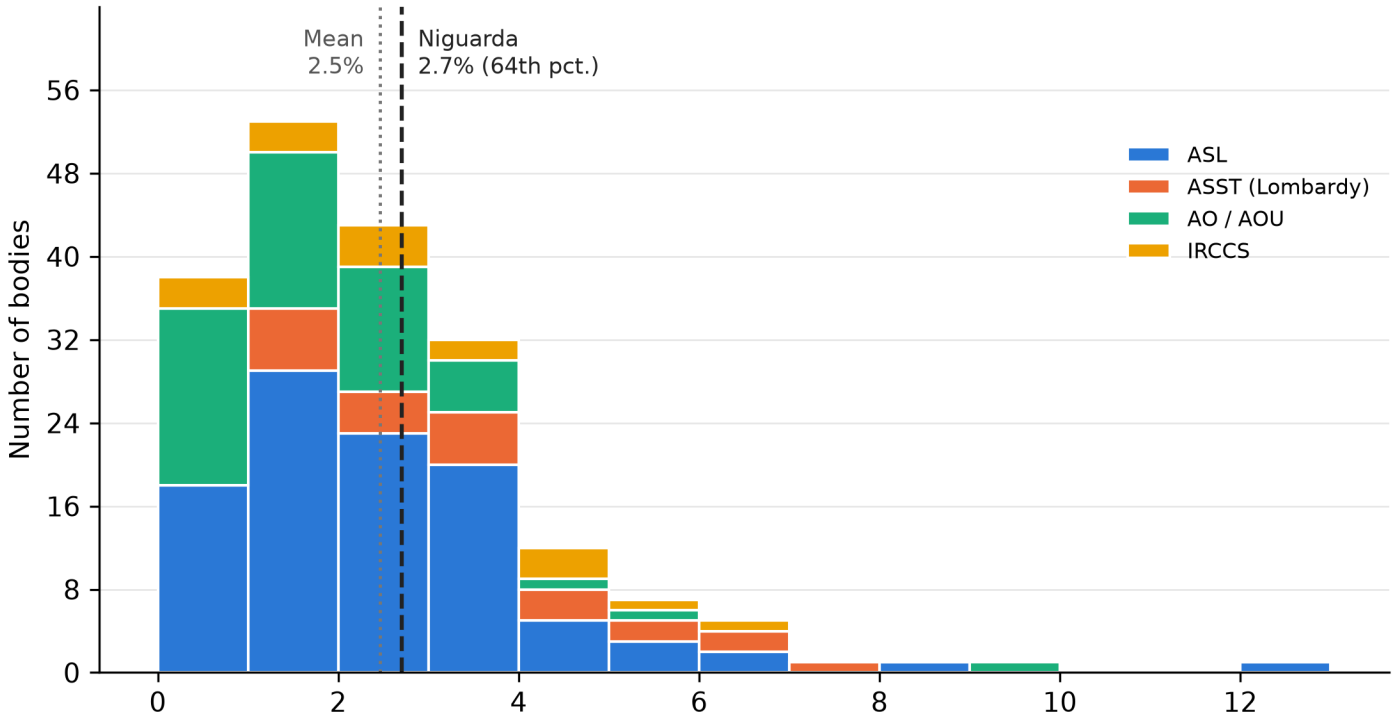
- traditional public employee
- fixed wage + career progression
- insider

Contrattista

- 2010-2018 freeze
- paid per shift + renewal risk
- outsider

Important: ED productivity not rewarded or even measured

Contractors Are Used Across Italian Hospitals — Niguarda Is Typical



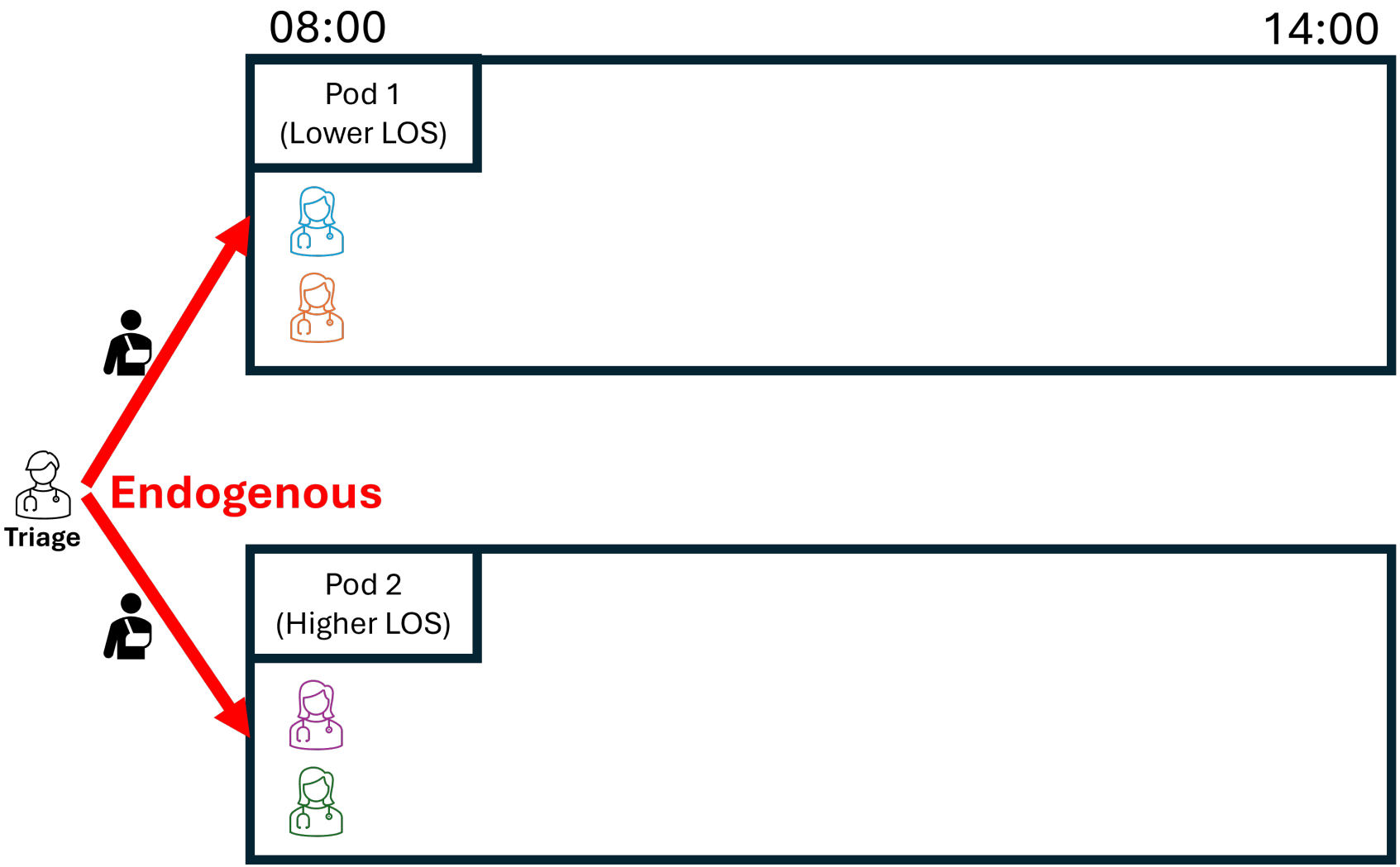
Share of contractors in total labour costs (%); N=194

→ more on this later

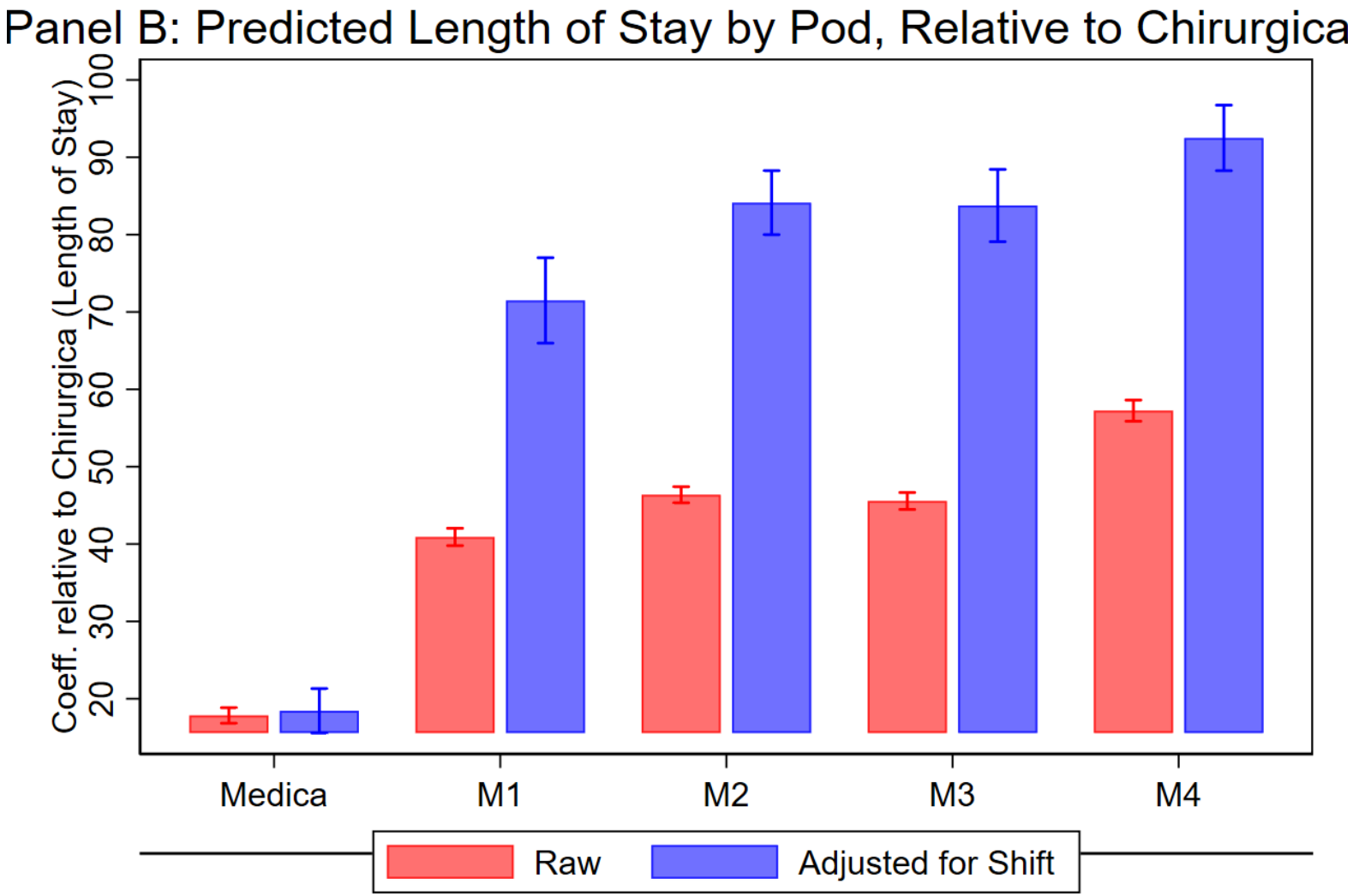
Doctors in Same Pod Work Under Same Conditions



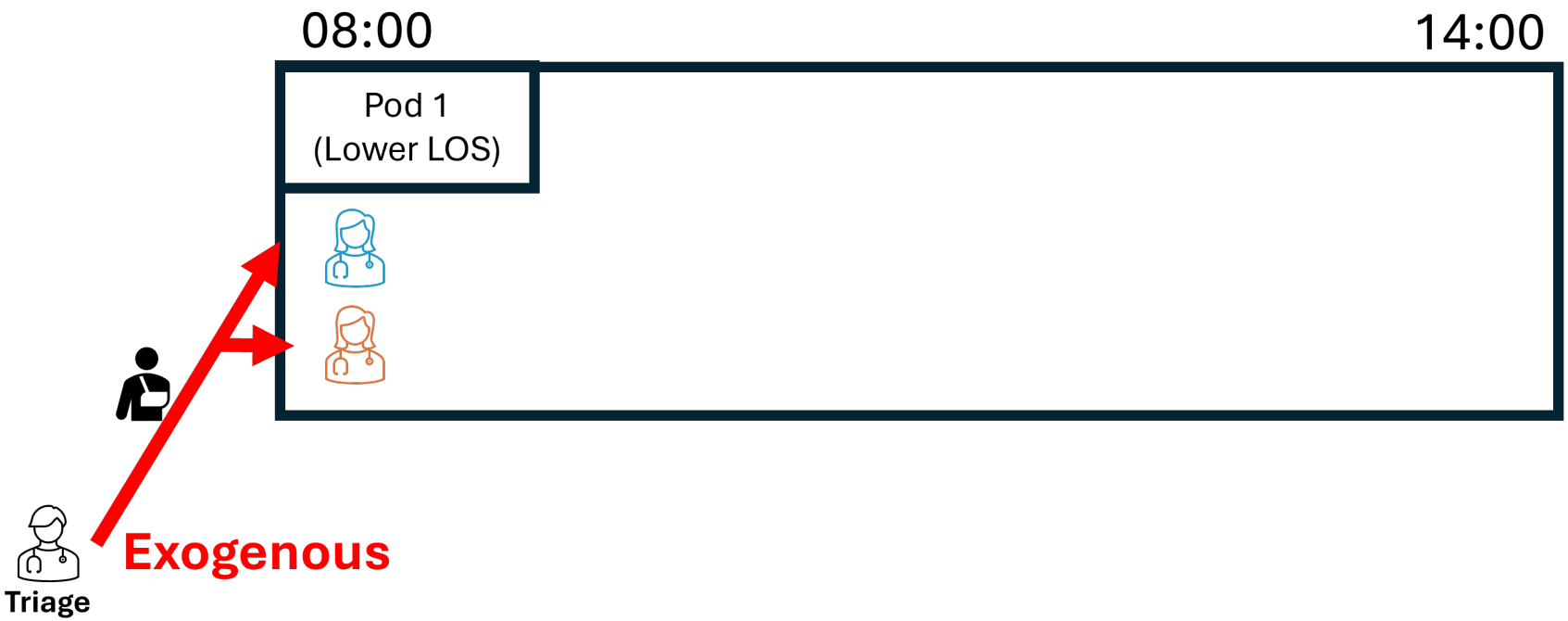
Assignment of Patients Across Pods



Predicted Length of Stay Is Different Across Pods



Assignment of Patients Within Pods



Why?

- nurse looks at existing caseloads, assesses whose caseload is lighter and assigns incoming patient to that doctor
- **important:** no match between patient characteristics and doctor characteristics
- **justification:** 'I was there' + Balance Tests + Other Stuff

Dependent Variables (e.g., Chan and Chen 2024)

- **Resource Use**:
  - **length of stay**
  - **cost of tests/treatments**
  - hospitalisation
  - handover to next shift
  - duplications
  - **overstay**
- **Patient Health**:
  - death
  - bounce back (return to ED)

Baseline Regression

An observation is a case/patient i:

$$y_i = \beta_1 \, contractor_{jt(i)} + \theta_{t(i)p(i)} + \epsilon_i$$

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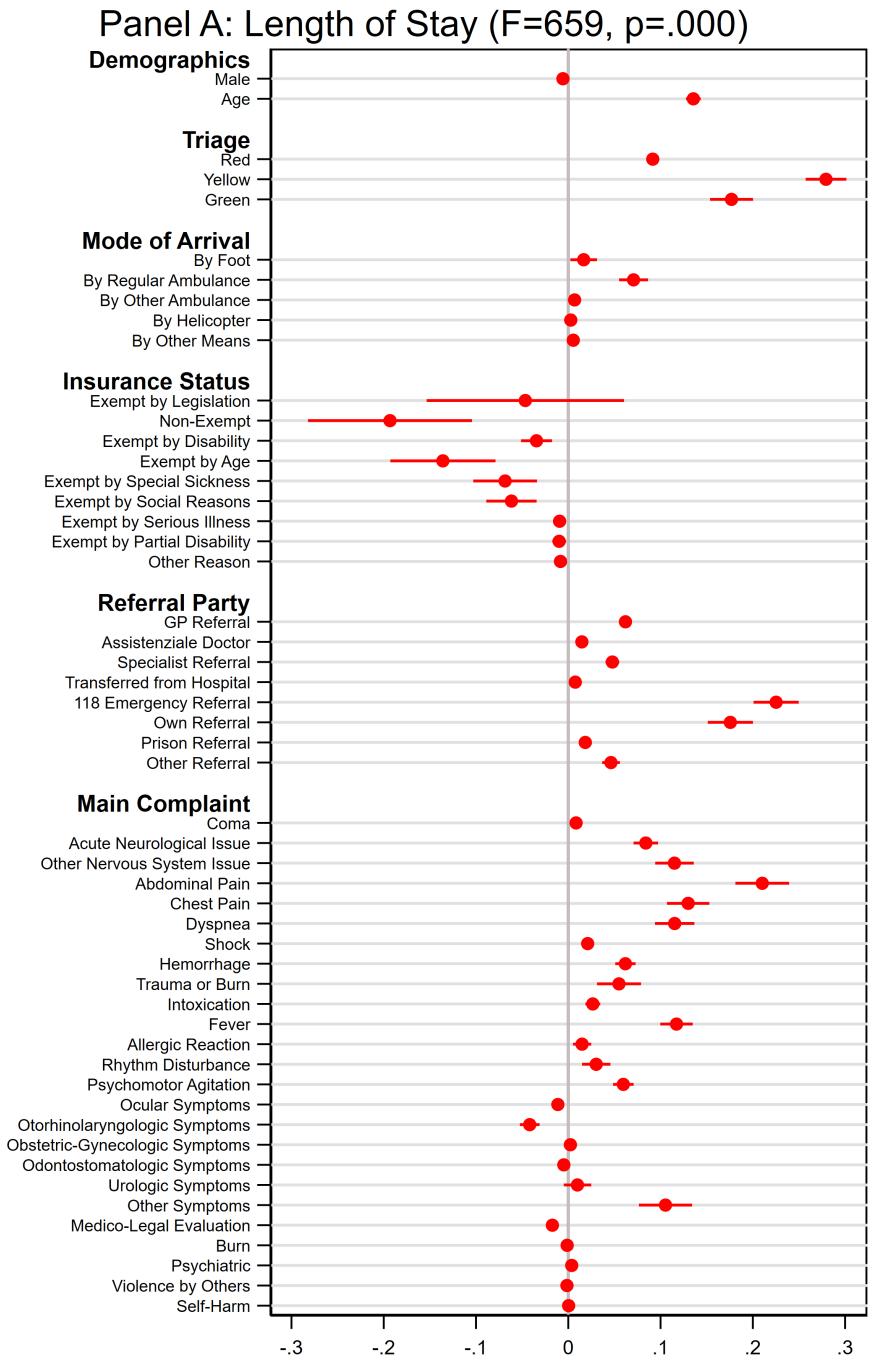
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- **Data:**
  - 2014-2023
  - 85 doctors
  - 300,000 cases

Patients Characteristics Strongly Predict Length of Stay

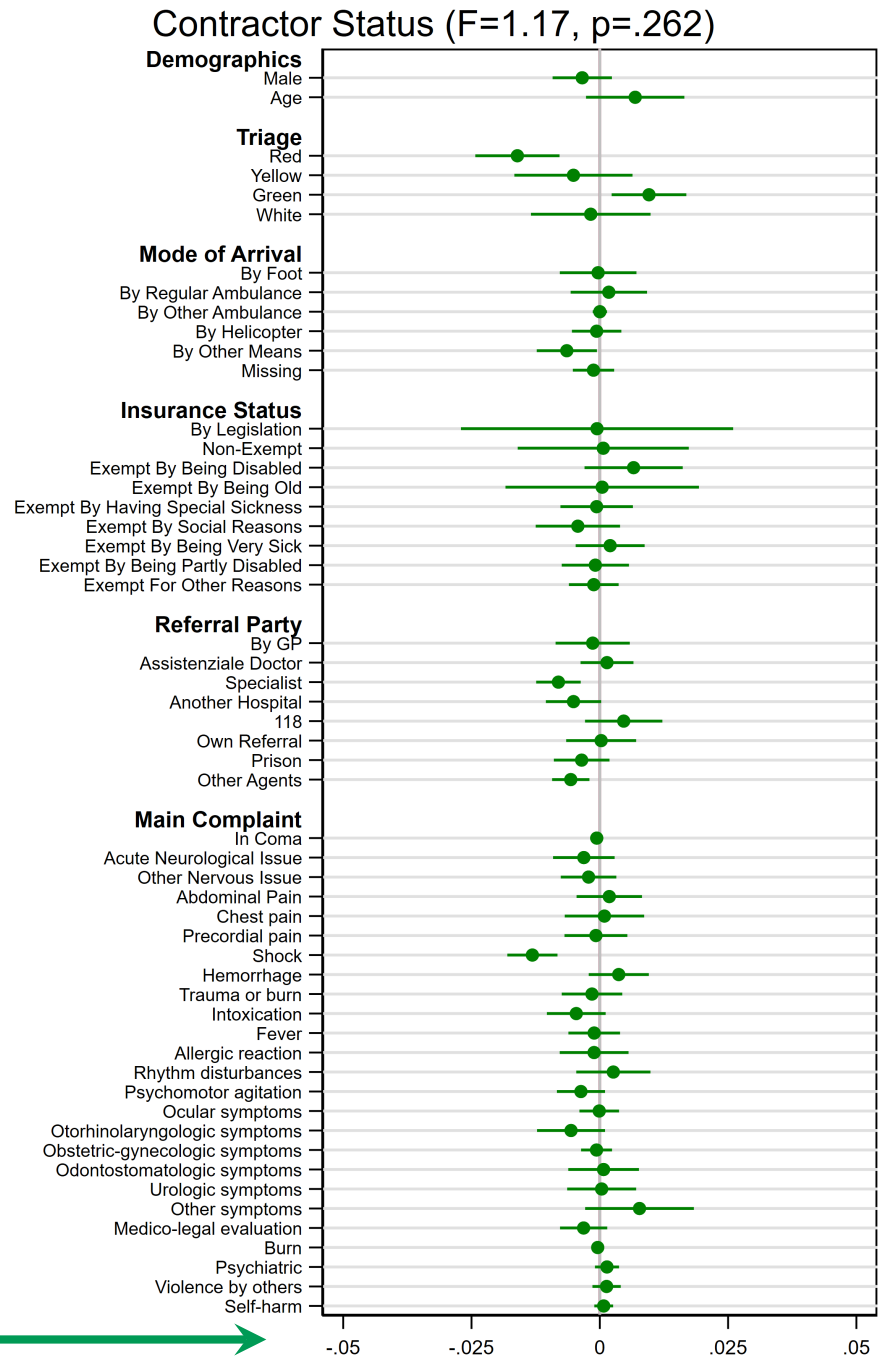
$$Length\,Stay_i = \delta X_i + \theta_{t(i)p(i)} + \epsilon_i$$



Patients Characteristics Uncorrelated with Contractor Assignment

$$Contractor_i = \gamma X_i + \theta_{t(i)p(i)} + \epsilon_i$$

Note the much smaller scale →



Result 1: Contractors use more resources

TABLE 2 - BASELINE RESULTS

	ED Cost	Length of Stay
Panel A: Controlling Only for Shift × Pod		
Contractor	.073***	.268***
	(.020)	(.036)
Panel B: Adding Patient Characteristics		
Contractor	.073***	.263***
	(.016)	(.039)
Panel C: Adding Exact Hour		
Contractor	.100***	.313***
	(.018)	(.048)

Costs and Length of Stay in logs. **Controls:** shift×pod FE, student doctor, patient characteristics; $N = 306{,}635$.

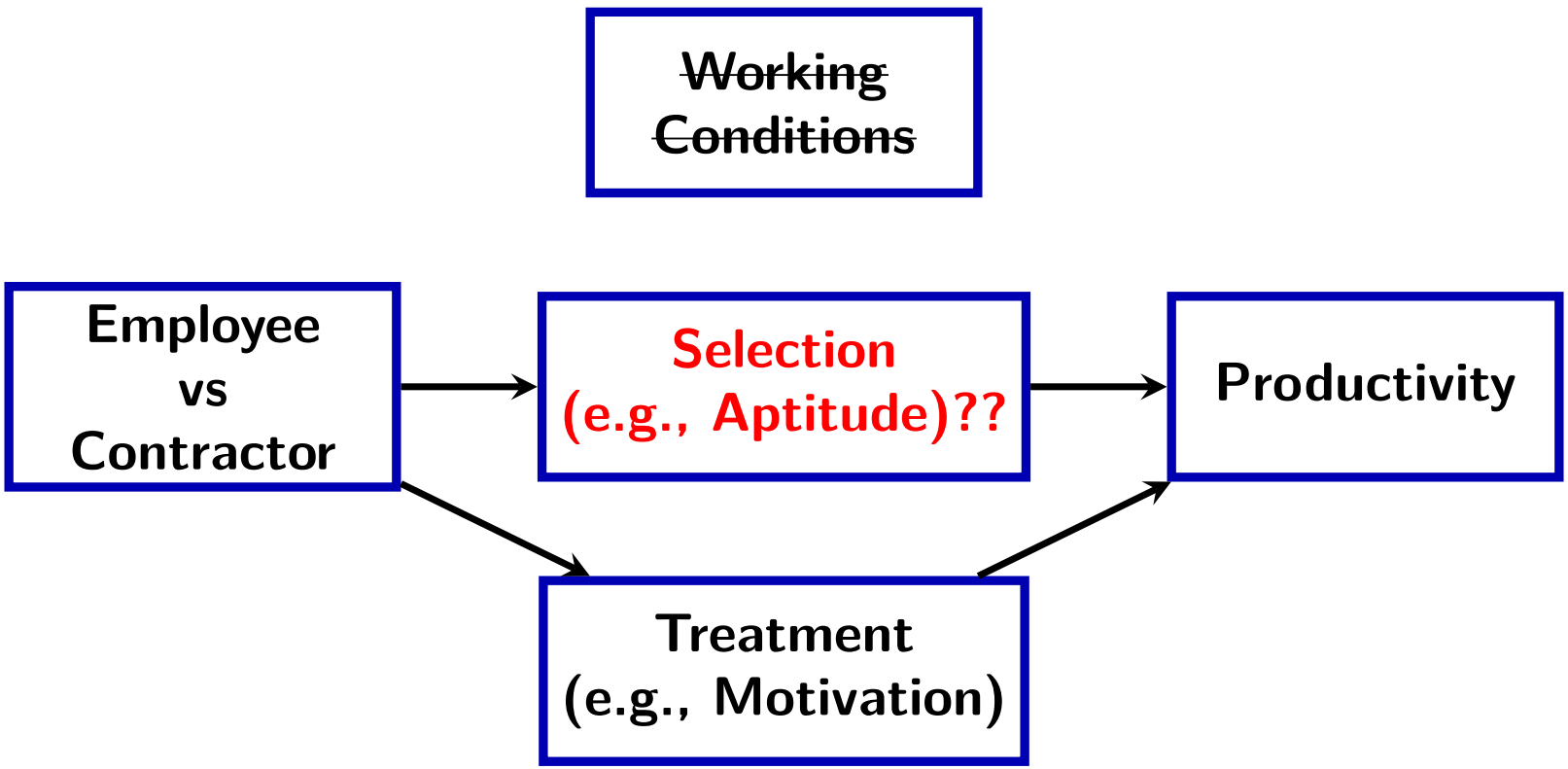
patient controls predicted vs actual within-shift controls rotation controls

Result 1: Contractors use more resources

Magnitudes

- $\approx$ **77 additional minutes** per patient or **1 SD** of doctor fixed effects
- $\approx$ **11 additional euros** per patient or **0.75 SD** of doctor fixed effects

Maybe Contractors Are Less Able?



We Have Within-Doctor Variation in Contracts

Observed in the data

- 14 of 18 contractors become employees (after 2016)

Institutional process

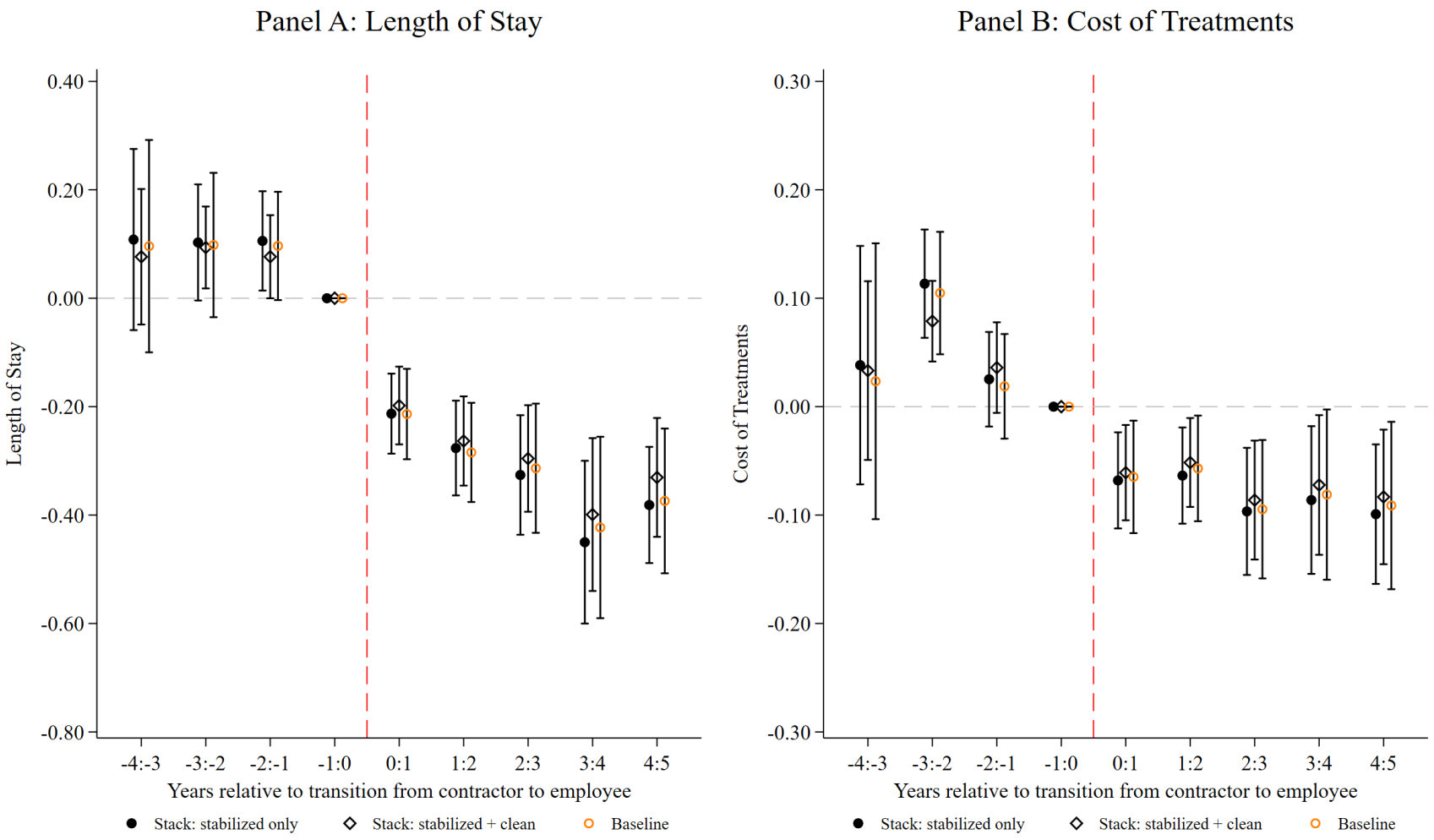
Vacancy opens → National competition → Objective ranking → Top-ranked candidate hired

Why this matters

- Ranking based on exam + CV + publications *rigid* criteria
- Niguarda has no discretion over who is hired
- No reference letters, prior performance, etc.

Baseline + doctor FE

Result 2: Discontinuously



Pre-trends test + Age / Experience Stabilised vs Clean Imputation estimator

Result 3: Contractors do not generate different health outcomes

OTHER PATIENT OUTCOMES

	Death in ED	Return to ED 2	Return to ED 5	Return to ED 30	Return to ED 60
Panel A: Without Doctor Fixed Effects					
Contractor	-.0003	.0008	-.0003	-.0006	-.0019
	(.0002)	(.0020)	(.0025)	(.0037)	(.0043)
Panel B: With Doctor Fixed Effects					
Contractor	-.0002	-.0004	-.0008	.0001	-.0002
	(.0005)	(.0033)	(.0047)	(.0062)	(.0070)
Mean Variable	.0008	.0364	.0650	.1328	.1741

Controls: shift×pod FE, student doctor, exact-hour FE, patient characteristics. SEs clustered by doctor and shift×pod. $N = 305{,}130$.

So Far:

1. contractors use more resources (e.g. by one S.D. of doctor fixed effects)
2. no differences in health outcomes
3. becoming an employee (discontinuously) decreases resource use

Other Results in the Paper - Contractors:

4. order more all kinds of things (e.g. blood tests, x-rays, ultrasounds...) →
5. are more likely to hand cases over to doctor(s) in next shift →
6. generate more duplications e.g., order same test several times →
7. do not generate spillovers on other doctors in same pod/shift →

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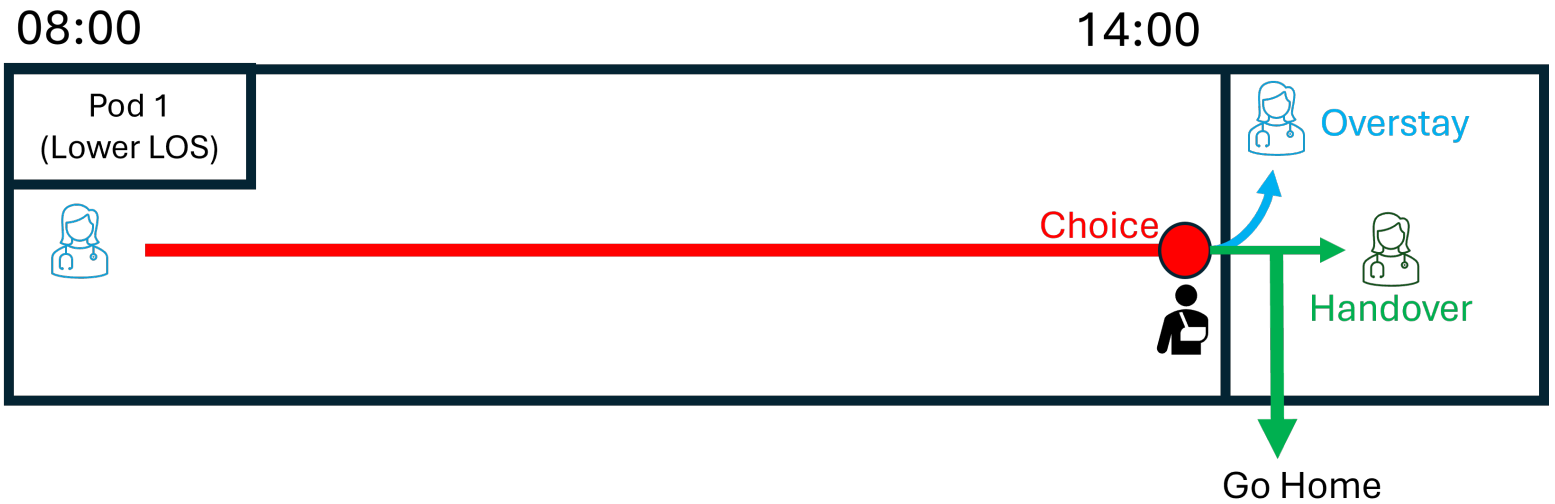
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Doctors Can Choose to Overstay



- doctors can leave when shift is over
- some doctors (4%) overstay to discharge patients and avoid handover
- no paid overtime
- **overstay suggestive of voluntary effort to help others**
  - → supporting this: overstay rises when the next doctor is similar

Result 4: Employees Overstay More Than Contractors

EFFECTS ON OVERSTAY VARIABLE

	(1)	(2)	(3)	(4)
Employee		**.019*** **	.014***	.013***
		(.003)	(.005)	(.005)
Diff. Gender	-.003*			-.003**
	(.002)			(.002)
Diff. Age / 10	-.005***			-.004***
	(.002)			(.002)
Doctor F.E.	Yes	No	Yes	Yes

Overstay = discharged by original doctor within 60 minutes after shift end. **Controls:** shift×pod FE and patient characteristics; Columns 2–4 add student doctor dummy. SEs clustered by doctor and shift×pod. Mean dep. var. = 0.04; $N = 305{,}169$.

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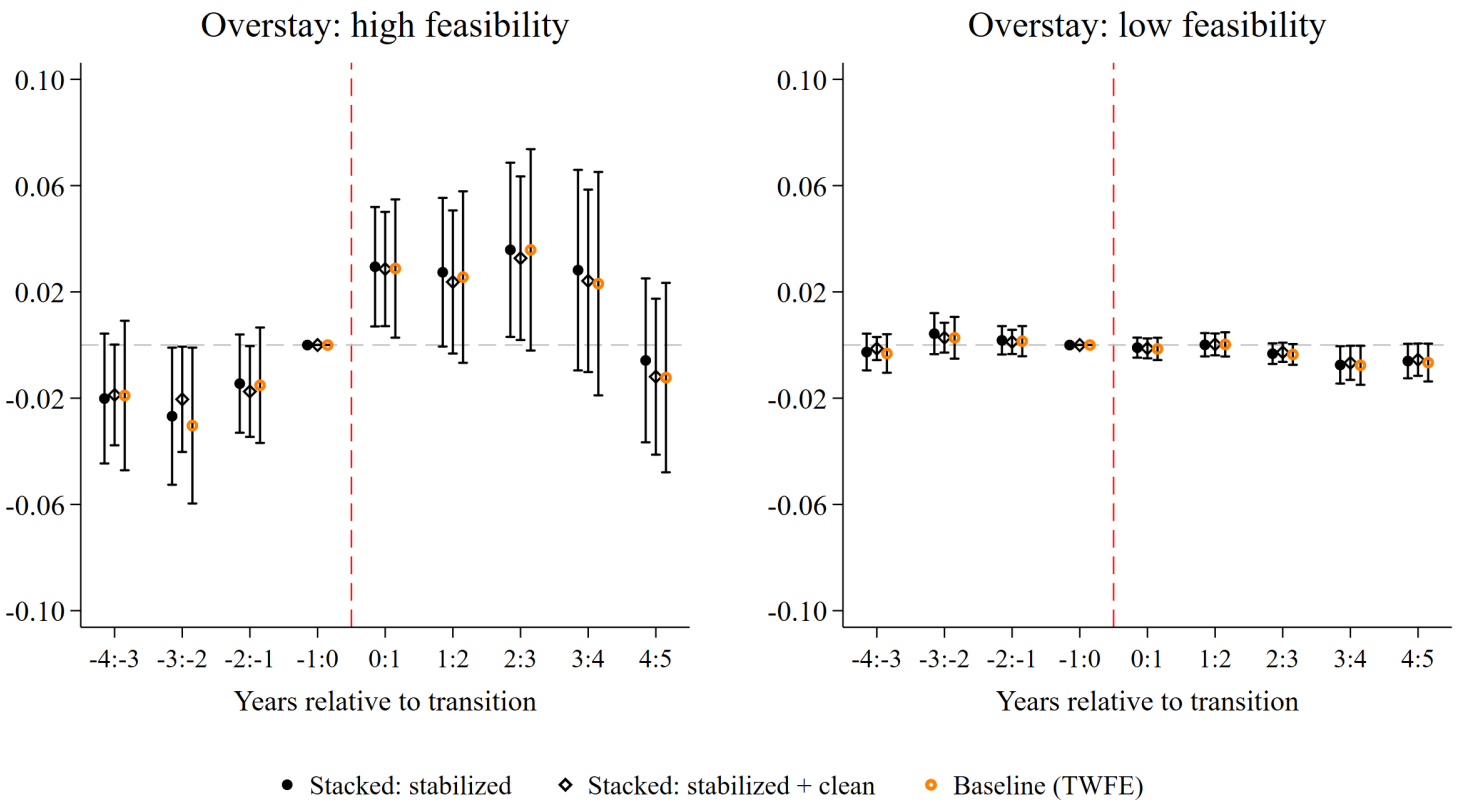
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Same doctor is more willing to overstay after transitioning to employee

Result 4: Employees Overstay More Than Contractors



High feasibility: cases predicted at assignment to be open at shift end but closable soon after

Effect only where staying can finish the case; low-feasibility half works as a placebo

Why Do Employees Overstay More?

To help the organisation

To help their colleagues

Something Else

- incentives?
- liability insurance?
- marginal cost of effort?

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- **maybe other employees are colleagues**
- **but contractors are not colleagues** → **overstay more if incoming doctor is employee**

Something Else

- incentives?
- liability insurance?
- marginal cost of effort?

who comes next should not matter

Result 5: Employees Overstay Only To Help Other Employees

EFFECTS ON OVERSTAY VARIABLE

	(1)	(2)	(3)
Employee	.005	.001	.002
	(.006)	(.006)	(.006)
Employee × Incoming Employees	.015***	.015***	.013**
	(.006)	(.006)	(.006)
Diff. Gender			-.003*
			(.002)
Diff. Age / 10			-.004***
			(.001)
P-value 1+2	.000	.001	.004
Doctor F.E.	No	Yes	Yes

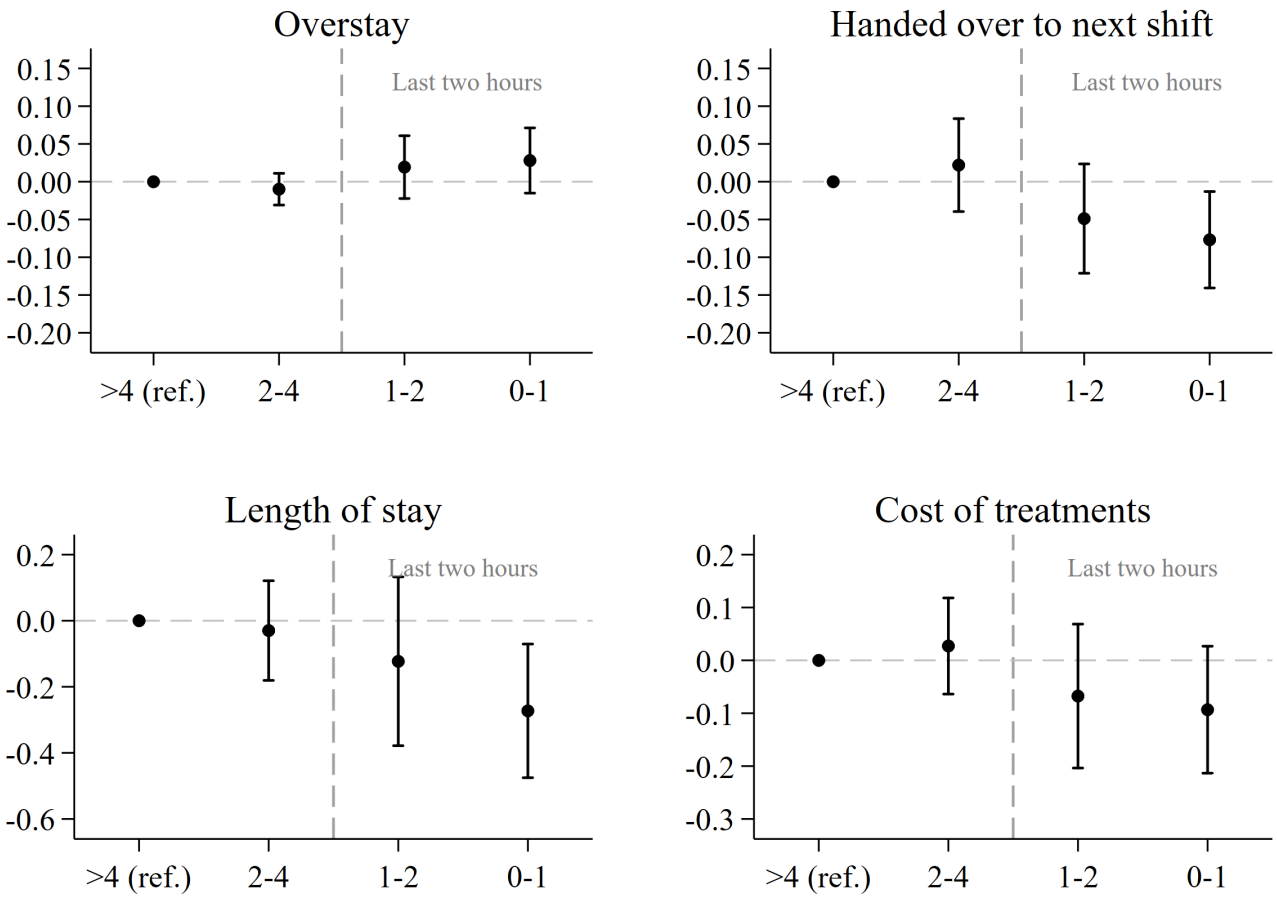
Overstay = discharged by original doctor within 60 minutes after next shift starts. Diff. gender/age and Incoming Employee compare original doctor to doctors in same pod next shift. **Controls:** shift×pod FE and patient characteristics, student doctor dummy. SEs clustered by doctor and shift×pod. Mean dep. var. = 0.04; $N = 305{,}094$.

LOS and ED Cost Also Lower When Employees Incoming

HETEROGENEITY BY EMPLOYEES FOLLOWING EMPLOYEE

	(1)	(2)	(3)	(4)
Panel A: Dependent Variable = Length of Stay				
Employee	-.323***	-.316***	-.201***	-.249***
	(.045)	(.032)	(.064)	(.048)
Employee × Incoming Employees			-.138***	-.077
			(.050)	(.047)
P-value 1+2			.000	.000
Doctor F.E.	No	Yes	No	Yes
Panel B: Dependent Variable = ED Cost				
Employee	-.105***	-.090***	-.059**	-.067***
	(.018)	(.018)	(.026)	(.027)
Employee × Incoming Employees			-.052**	-.026
			(.023)	(.021)
P-value 1+2			.000	.000
Doctor F.E.	No	Yes	No	Yes

Effect of Employee × Incoming Employees Within the Shift



→ Effects concentrate on last two hours of the shift

Mechanism: Becoming an Employee Increases Willingness to Cooperate with Employees

Social Identity:

- employment status shapes who you see as "us" (Akerlof and Kranton, 2010)
- insiders internalise the welfare of other insiders

Relational Contracts:

- cooperation norm within the employee community (exclude contractors)
- switches on for the *same doctor* when they cross the boundary
- weak evidence for *one-to-one* favour exchange →

Gibbons and Henderson (2012): high-effort norms, organisational culture... can be understood through the lens of relational contracts

Beyond Niguarda: Lombard Hospitals

- **Data:** 24 million ED visits in Lombardy
- Same outcomes (**length of stay**, **ED cost**) for **61 hospitals**

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  - not ED-specific (whole hospital)
  - lots of measurement error

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Still worth checking: do hospitals that rely more on contractors have longer stays and higher costs?

Beyond Niguarda: Evidence from Lombard Hospitals

Regress outcomes on hospital-wide share of contractors

	Length of Stay	ED Cost
	(+1pp contractor share)	
Cross-Sectional	.055**	.014*
(patient + hospital controls)	(.019)	(.008)
Hospital-Year Panel	.034*	.010*
(hospital FE + year FE)	(.017)	(.006)

Outcomes in logs; controls for patient and hospital characteristics.
SEs clustered by health trust. 61 hospitals, 626 hospital-years.

More reliance on contractors ⇒ longer stays, higher cost

Contributions

1. **first study** of productivity differences contractors and employees
2. **disentangles** selection and treatment
3. provides evidence that employee status **motivates through willingness to cooperate with other employees**

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External validity matters: But proof of concept that contractor status affects productivity independently of selection and working conditions

Additional Slides

Robustness: Main Result

- → Controlling for Within-Shift Assignment
- → Controlling for Recent Rotation Patterns
- → Robustness to Patient Controls
- → Predicted vs Actual Length of Stay
- → Within-Doctor Variation

Event Study & Pre-Trends

- → Testing Pre-Trends Using Only the Pre-Period
- → Event Study + Age and Experience Controls
- → Stabilised Employees vs. Clean Controls
- → Imputation Estimator

Other Results

- → Contractors Really Like Ordering Tests
- → Contractors Hand Over More and Leave More Cases Open
- → Handovers Associated with More Duplication
- → Employees Do Not Create Spillovers
- → Contractors Are Assigned Less Cases

Mechanisms: Identity vs. Relational Contracts

- → Overlap with Same Doctors
- → Does Past/Future Overlap Predict Overstay?
- → Controlling for Overlap Patterns

Appendix: Controlling for Within-Shift Assignment

TABLE A3
BASELINE ESTIMATES
CONTROLLING FOR WITHIN-SHIFT ASSIGNMENT VARIABLES

	ED Cost			Length of Stay		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: No Doctor Fixed Effects						
Contractor	.105***	.105***	.096***	.323***	.323***	.304***
	(.018)	(.018)	(.018)	(.045)	(.045)	(.043)
Hours to End of Shift		.008	-.011*		.004	-.034***
		(.007)	(.006)		(.013)	(.012)
Cases Previously Assigned in Shift			-.011***			-.023***
			(.001)			(.003)
Panel B: With Doctor Fixed Effects						
Contractor	.090***	.085***	.085***	.316***	.304***	.304***
	(.018)	(.018)	(.018)	(.032)	(.032)	(.032)
Hours to End of Shift		-.005	-.005		-.023*	-.023*
		(.006)	(.006)		(.012)	(.012)
Cases Previously Assigned in Shift			-.007***			-.017***
			(.001)			(.003)

Outcomes in logs. **Controls:** shift×pod FE and patient characteristics; Panel B adds doctor FE and exact-hour FE. Columns 2/5 add hours to shift end; Columns 3/6 add cases previously assigned in shift. SEs clustered by doctor and shift×pod. $N = 305{,}205$.

Back

Appendix: Controlling for Recent Rotation Patterns

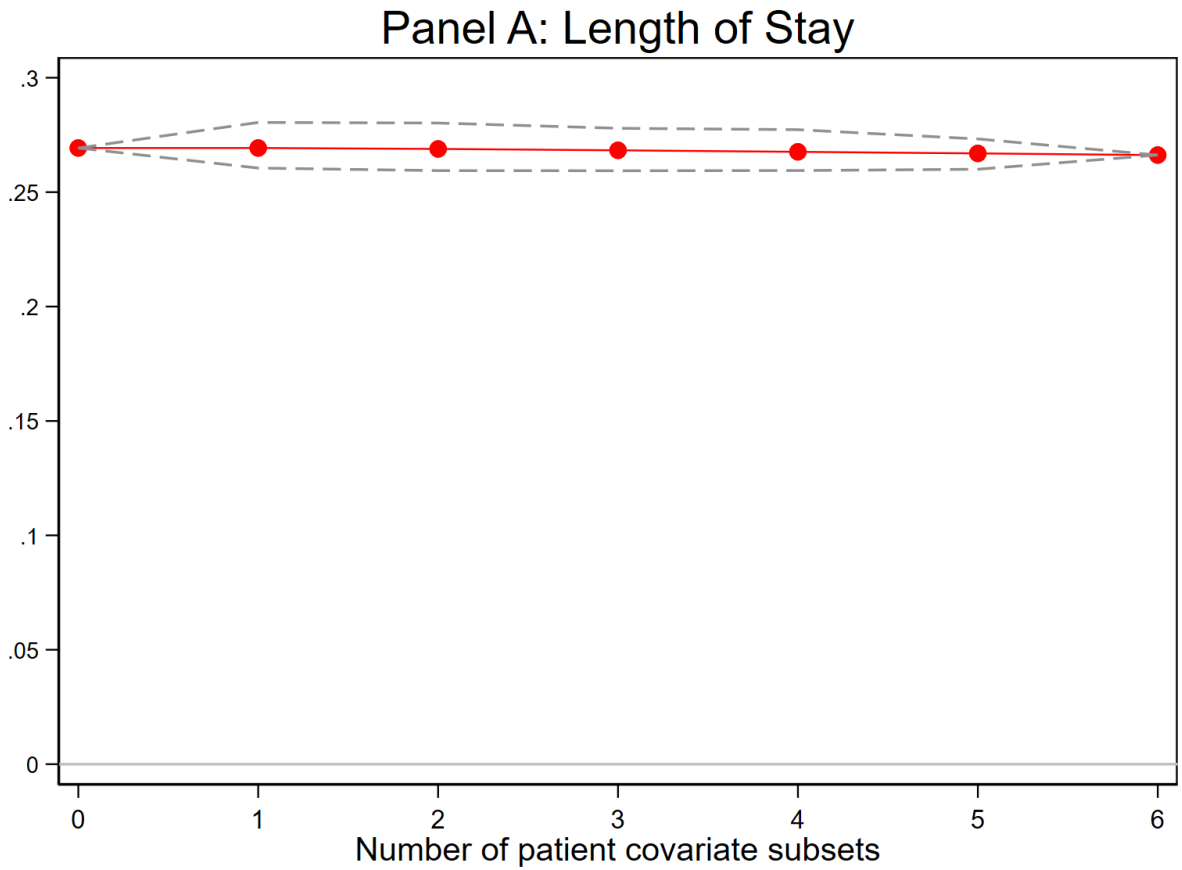
TABLE A5
BASELINE CONTROLLING FOR RECENT ROTATION PATTERNS

	ED Cost		Length of Stay	
	(1)	(2)	(3)	(4)
Contractor	.090***	.090***	.316***	.290***
	(.018)	(.018)	(.032)	(.032)
Total Shifts		-.005		-.013
		(.005)		(.011)
Morning Shifts		-.001		.006
		(.003)		(.005)
Afternoon Shifts		.002		.011*
		(.003)		(.006)
M1 Shifts		.003		-.006
		(.004)		(.008)
M2 Shifts		-.001		-.011
		(.004)		(.007)
M3 Shifts		.001		-.008
		(.004)		(.007)
M4 Shifts		-.001		-.007
		(.005)		(.008)
Medica Shifts		.005		.009
		(.003)		(.006)
Monday Shifts		.001		-.002
		(.004)		(.010)
Tuesday Shifts		.007		.016
		(.005)		(.011)
Wednesday Shifts		.002		.001
		(.005)		(.011)
Thursday Shifts		.002		.002
		(.005)		(.011)
Friday Shifts		.001		.008
		(.005)		(.011)
Saturday Shifts		.008		.013
		(.006)		(.011)
Doctor Fixed Effects	Yes	Yes	Yes	Yes

Back

Appendix: Additional Test of Identification

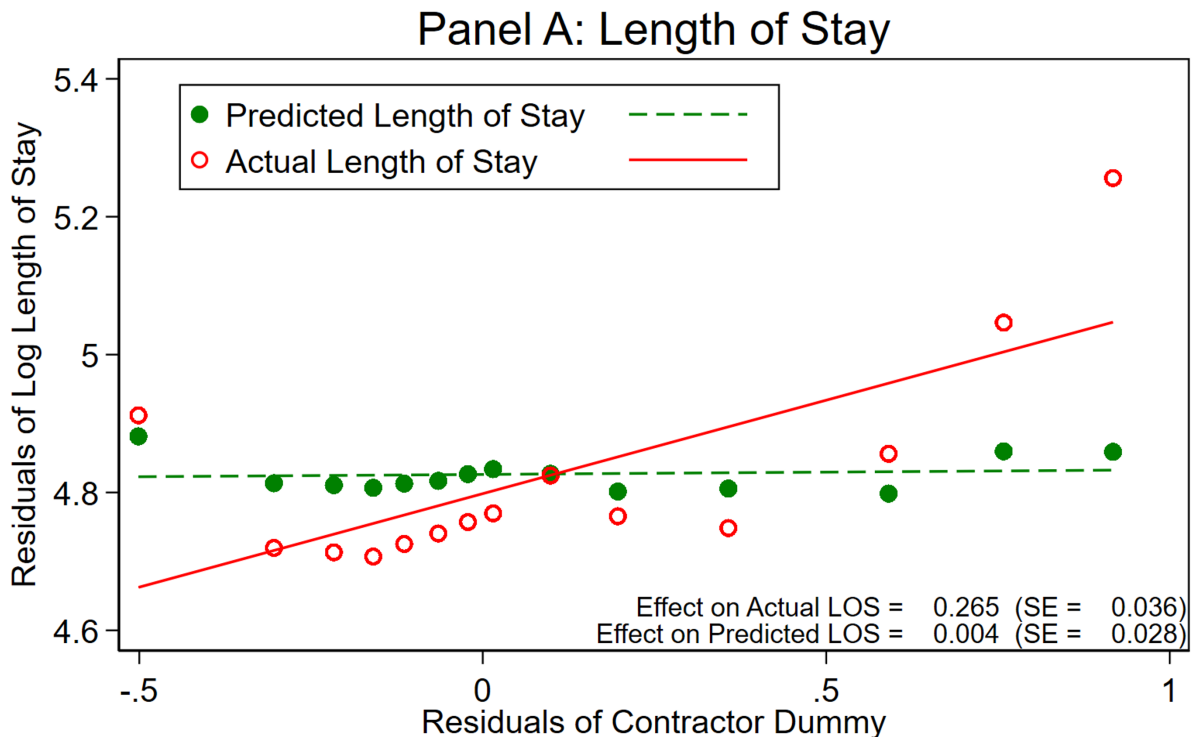
Robustness to adding 7 sets of controls: gender, age, triage, main complaint, referral source, mode of arrival, insurance status



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Appendix: Additional Test of Identification Number 2

Plotting predicted (on the basis of observables) and actual length of stay as a function of contractor dummy.



(Both axes residualised from shift × pod)

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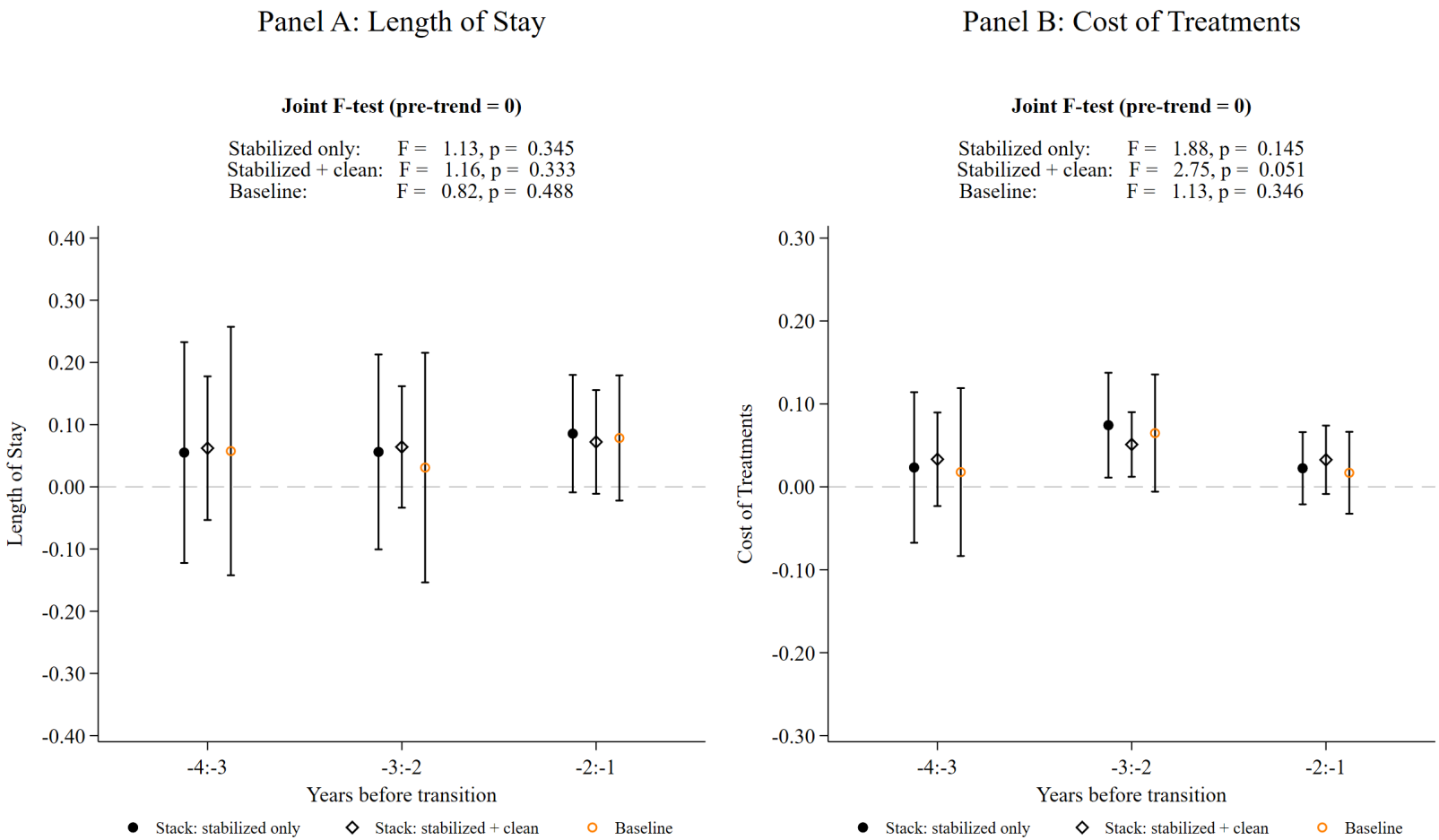
Within-Doctor Variation

BASELINE RESULTS (Cont)

	ED Cost	Length of Stay
Panel D: Adding Doctor Fixed Effects		
Contractor	.090***	.316***
	(.018)	(.032)
Panel E: Adding Doctor Characteristics		
Contractor	.100***	.236***
	(.020)	(.038)
Log Doctor Age	-.280	-2.356
	(.733)	(1.465)
Log Doctor Past Cases	-.003	-.016
	(.007)	(.014)
Log Doctor Tenure at Niguarda	.016*	-.013
	(.009)	(.016)

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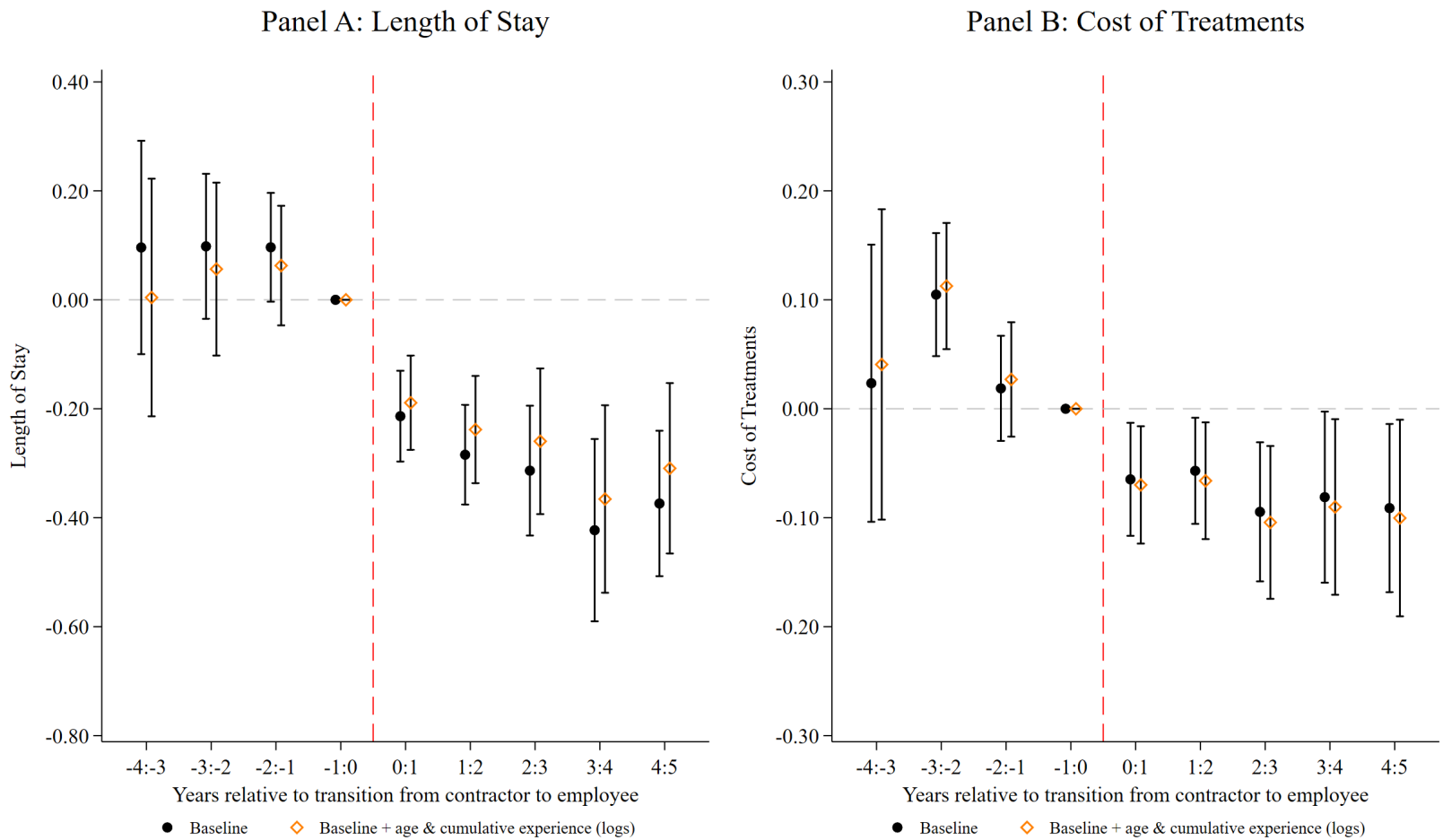
Appendix: Testing Pre-Trends Using Only the Pre-Period



When estimated on the pre-period only, lead coefficients are small and jointly insignificant.

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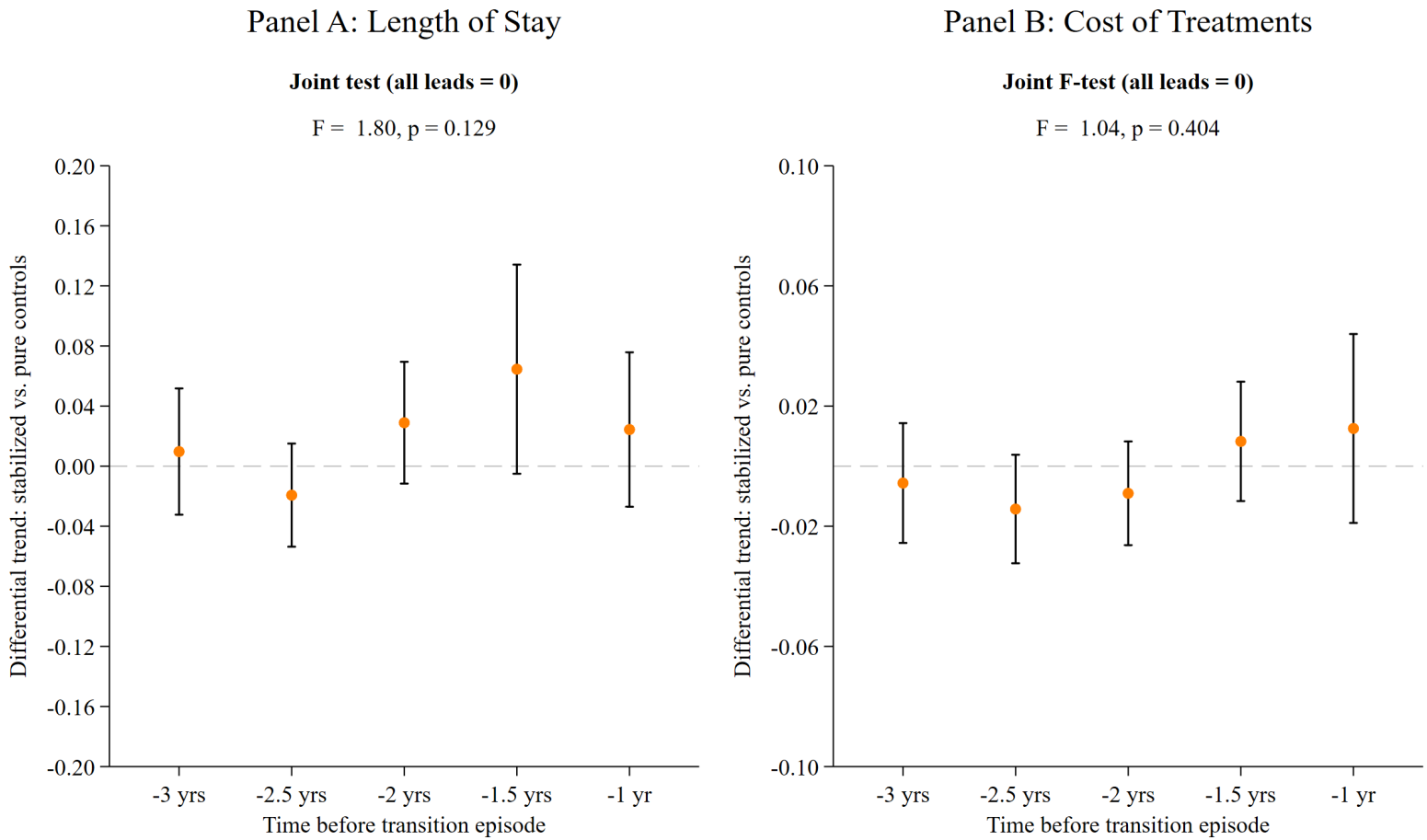
Appendix: Event Study + Age and Experience Controls



Adding controls for doctor age and experiences does not change dynamic profile.

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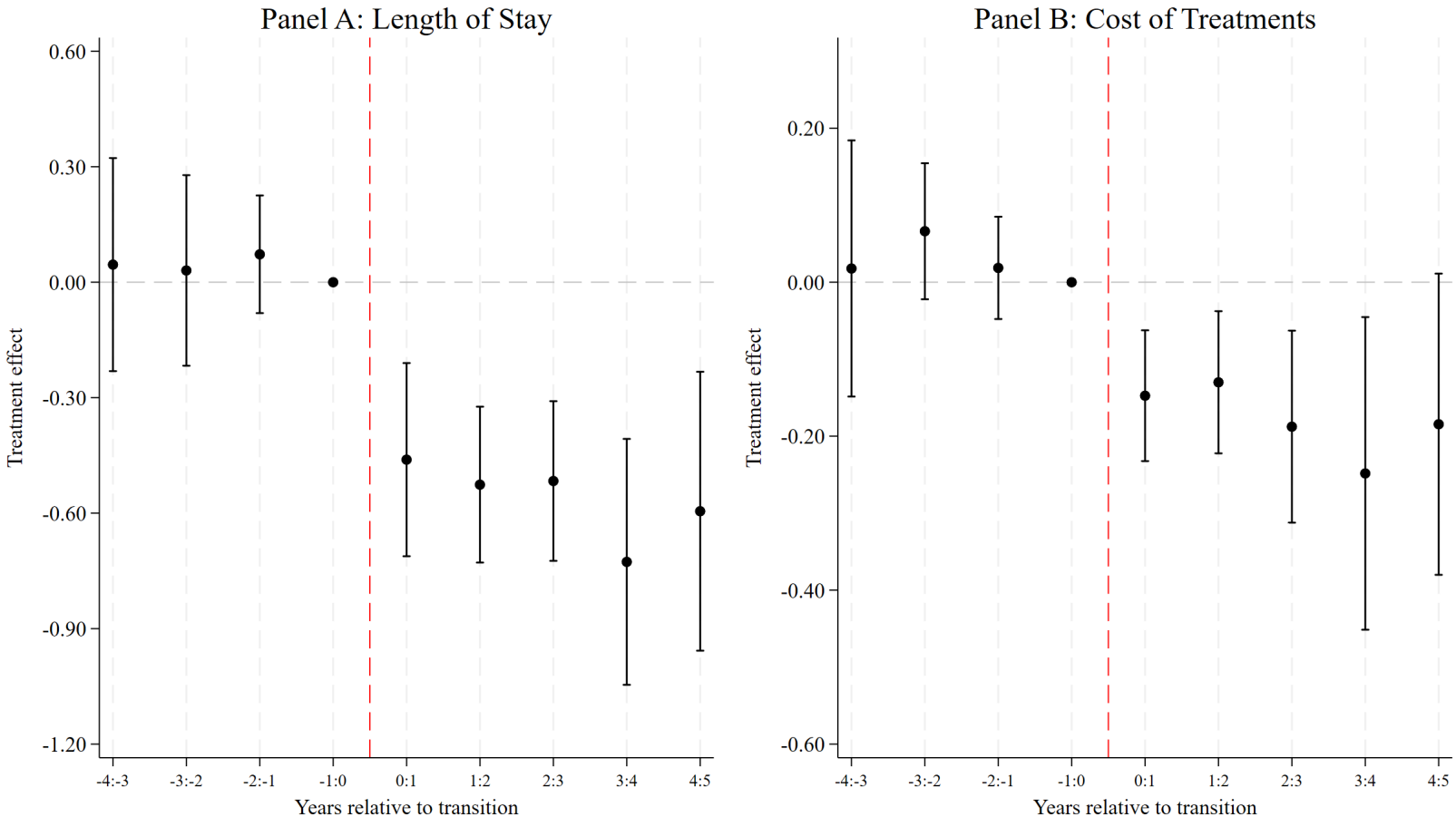
Appendix: Stabilised Employees vs Clean Controls



In pre-transition windows, stabilised employees move very similarly to the clean control group.

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Appendix: Imputation Estimator



Pre bins: separate pre-trend regression; post bins: imputation estimator. Bootstrap SEs (100 replications, doctor-level resampling).

A modified imputation estimator delivers same qualitative pattern: small pre-trends and persistent post-transition declines.

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Contractors Really Like Ordering Tests

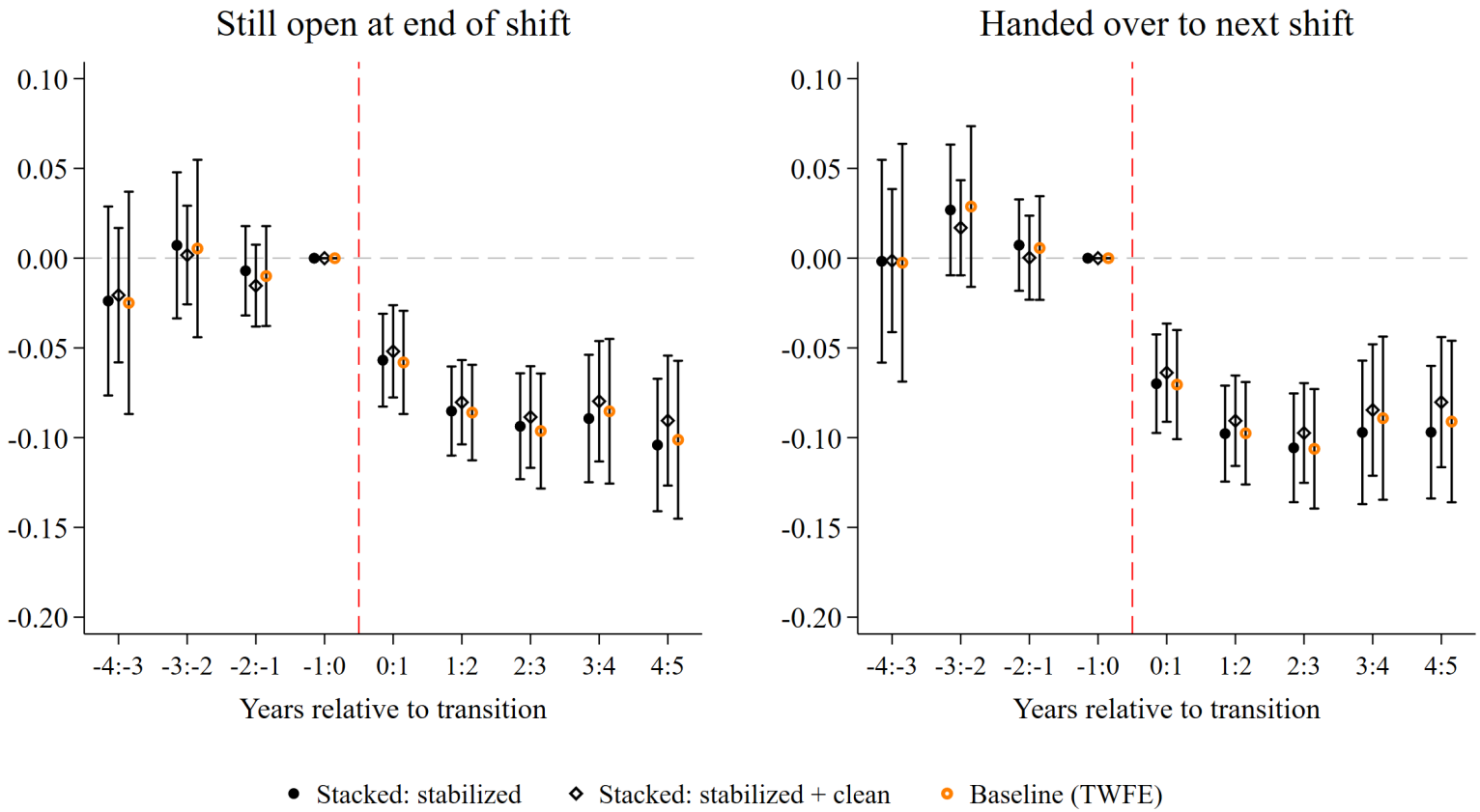
EFFECTS ON SELECTED COST ITEMS

	COST TYPES		SELECTED TESTS				
	Cost of Laboratory Tests	Other Costs	Blood Test	X-Ray	CT Scan	Electro-Cardiogram	Ultra-Sound
Panel A: Not Controlling for Doctor Fixed Effects							
Contractor	.457***	.096***	.06***	.1***	.01	.024*	.013**
	(.065)	(.018)	(.007)	(.014)	(.007)	(.014)	(.007)
Panel B: Adding Doctor Fixed Effects							
Contractor	.393***	.081***	.044***	.072***	.005	.039**	.016
	(.072)	(.022)	(.009)	(.017)	(.011)	(.017)	(.013)
Mean DV	Logs	Logs	.59	.62	.26	.28	.1

The cost of laboratory tests and the other costs variables are in logs. All regressions control for whether the assigned doctor is a student, the interaction of shift (i.e. the morning, afternoon or night shift within a specific date) and pod, exact hour and patient characteristics (age, gender, triage, main complaint, mode of arrival and referral party). Standard errors two-way clustered at the doctor and shift-by-pod interaction level. The number of observations is 305,205.

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Contractors Hand Over More. . . and Leave More Cases Open



Same doctor, after becoming an employee: fewer cases left open at end of shift and fewer handovers.

handover table Back

Contractors Hand Over More to Next Doctor

EFFECTS ON CONSULTATIONS, HANDOVERS AND DUPLICATED COSTS

	(1) Required Specialist Consultation	(2) Handover to Next Shift	(3) Total Duplicated Costs	(4) Share Duplicated Costs
Panel A: Not Controlling for Doctor Fixed Effects				
Contractor	.014***	.074***	.164***	.006***
	(.005)	(.011)	(.037)	(.001)
Panel B: Adding Doctor Fixed Effects				
Contractor	.012	.077***	.222***	.009***
	(.009)	(.012)	(.05)	(.002)
Mean Dep.	.35	.35	.71	.03

Standard errors two-way clustered at the doctor and shift-by-pod interaction level. All regressions control for whether the assigned doctor is the student, the interaction of shift (i.e. the morning, afternoon or night shift within a specific date) and pod, exact hour and patient characteristics (age, gender, triage, main complaint, mode of arrival, referral party and insurance status). The number of observations is 305,130.

handovers vs. duplications Back

Contractors Create more Duplications

EFFECTS ON CONSULTATIONS, HANDOVERS AND DUPLICATED COSTS

	(1) Required Specialist Consultation	(2) Handover to Next Shift	(3) Total Duplicated Costs	(4) Share Duplicated Costs
Panel A: Not Controlling for Doctor Fixed Effects				
Contractor	.014***	.074***	.164***	.006***
	(.005)	(.011)	(.037)	(.001)
Panel B: Adding Doctor Fixed Effects				
Contractor	.012	.077***	.222***	.009***
	(.009)	(.012)	(.05)	(.002)
Mean Dep.	.35	.35	.71	.03

Standard errors two-way clustered at the doctor and shift-by-pod interaction level. All regressions control for whether the assigned doctor is the student, the interaction of shift (i.e. the morning, afternoon or night shift within a specific date) and pod, exact hour and patient characteristics (age, gender, triage, main complaint, mode of arrival, referral party and insurance status). The number of observations is 305,130.

handovers vs. duplications Back

Handovers associated with more duplication

CORRELATIONS BETWEEN HANDOVERS TO THE NEXT SHIFT AND DUPLICATED COST ITEMS

	Log Number of Duplicated Items	Share of Duplicated Items
Handover to Next Shift	.291***	.031***
	(.008)	(.001)
Length of Stay	.078***	.010***
	(.004)	(.000)

Duplicated items: number and share of repeated cost items within an incident. Length of stay in logs. **Controls:** shift×pod FE, exact-hour FE, and patient characteristics. SEs clustered by doctor and shift×pod. Mean duplicated items = 0.71; mean share = 0.03; $N = 305{,}130$.

Even after controlling for LOS and patient characteristics, handover associated with ≈30% more duplicated cost items

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Employees do not behave differently when sharing with contractors

Use only observations from Employees:

$$y_i = \phi_1 \, number_of_contractors_{jt(i)} + \lambda_{t(i)} + \mu_{p(i)} + \epsilon_i$$

SPILLOVERS OF CONTRACTORS ON EMPLOYEES' PRODUCTIVITY

	ED Cost	Length of Stay
Panel A: Controlling Only for Shift		
Number of Contractors in Pod	.002	-.017
	(.007)	(.017)
Panel B: Adding Patient Characteristics		
Number of Contractors in Pod	-.001	-.019
	(.007)	(.016)
Panel C: Adding Doctor Fixed Effects		
Number of Contractors in Pod	-.003	-.020
	(.006)	(.014)

Employee observations only. Length of stay and ED cost in logs. **Controls:** number of doctors, number of students, shift FE, pod FE; SEs clustered by doctor and shift×pod. $N = 261{,}225$.

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Contractors are assigned less cases

Triage nurse assigns cases depending on outstanding queue

Sample of Doctors/Shifts:

$$cases_{jt} = \beta_1 contractor_{jt} + \beta_2 student_{jt} + \theta_{tp} + \epsilon_{jt}$$

EFFECTS ON NUMBER OF CASES DURING THE SHIFT

	Unweighted	Weighted by LOS	Weighted by ED Cost
Panel A: Controlling Only for shift-by-pod			
Contractor	-.078***	-.101***	-.103***
	(.025)	(.030)	(.028)
Panel B: Adding Doctors Fixed Effects			
Contractor	-.172***	-.208***	-.204***
	(.020)	(.025)	(.025)

Shift-by-pod doctor-level regressions. Outcomes: log assigned cases, or log cases weighted by predicted length of stay/cost. SEs clustered by doctor and shift×pod. $N = 81{,}681$.

Appendix: Doctors Are More Willing to Overstay for Others Similar to Them

TABLE 7: HOMOPHILY AND OVERSTAY

	Overstay
Gender differences with incoming doctors	-.003*
	(.002)
Age difference with incoming doctors / 10	-.005***
	(.002)
Doctor F.E.	Yes
Incoming Doctors F.E.	Included in shift-by-pod

Overstay = discharged by original doctor within 60 minutes after shift end. Differences in gender/age compare original doctor to doctors in same pod next shift. **Controls:** shift×pod FE and patient characteristics. SEs clustered by doctor and shift×pod. Mean dep. var. = 0.04; $N = 305{,}094$.

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Appendix: Overlap with Same Doctors

Employees are not more likely to overlap with the same doctors in the same shift, previous shift, or next shift.

TABLE A16
CORRELATION BETWEEN OVERLAP MEASURES AND EMPLOYEE STATUS

	Horizontal Overlap (Past)	Backward Handover (Past)	Forward Handover (Past)	Horizontal Overlap (Future)	Backward Handover (Future)	Forward Handover (Future)
Panel A: Without Doctor Fixed Effects						
Employee	-.009*	.002	.000	-.010**	.006***	.001
	(.005)	(.004)	(.002)	(.004)	(.002)	(.002)
Panel B: With Doctor Fixed Effects						
Employee	-.001	-.003	-.004*	-.001	.003	-.001
	(.004)	(.004)	(.002)	(.002)	(.002)	(.001)
Mean Variable	.16	.053	.053	.16	.047	.047

Regressions of overlap measures on employee status. Past/future variables measured over 365-day windows; horizontal overlap captures shared shifts, backward/forward handover captures whether doctors followed each other in the same pod. **Controls:** student doctor, shift×pod FE, exact-hour FE, patient characteristics. SEs clustered by doctor and shift×pod. $N = 302{,}166$.

Appendix: Does Past/Future Overlap Predict Overstay?

Doctors do not overstay more when same-shift, incoming, or previous-shift doctors are doctors with whom they overlap more often.

TABLE A17
OVERLAP VARIABLES AND OVERSTAY

	depvar = overstay	
Horizontal Overlap (Past)	.030	.022
	(.020)	(.018)
Backward Handover (Past)	-.090***	-.058***
	(.026)	(.024)
Forward Handover (Past)	-.032	-.012
	(.033)	(.031)
Horizontal Overlap (Future)	.030	.027*
	(.021)	(.016)
Backward Handover (Future)	-.083*	-.030
	(.043)	(.034)
Forward Handover (Future)	.028	.049
	(.036)	(.033)
Doctor F.E.	No	Yes

Regressions of overstay on overlap measures. Past/future variables measured over 365-day windows; horizontal overlap captures shared shifts, backward/forward handover captures whether doctors followed each other in the same pod. **Controls:** shift×pod FE, exact-hour FE, and patient characteristics. SEs clustered by doctor and shift×pod. $N = 286{,}330$.

Appendix: Controlling for Overlap Patterns

Controlling for overlap patterns leaves the baseline estimates for overstay unchanged.

TABLE A18
EFFECTS ON OVERSTAY VARIABLE CONTROLLING FOR OVERLAP VARIABLES

	depvar = overstay		
Employee	.006	.001	.002
	(.006)	(.006)	(.006)
Employee × Incoming Employees	.017***	.016***	.014**
	(.006)	(.006)	(.006)
Difference in Gender			-.003
			(.002)
Difference in Age / 10			-.003*
			(.002)
P-value 1+2	.000	.000	.001
Doctor Fixed Effects	No	Yes	Yes
Past Overlap Variables	Yes	Yes	Yes
Future Overlap Variables	Yes	Yes	Yes

Overstay = discharged by original doctor within 60 minutes after shift end. Diff. gender/age and Incoming Employee compare original doctor to doctors in same pod next shift. **Controls:** student doctor, shift×pod FE, and patient characteristics. SEs clustered by doctor and shift×pod. Mean dep. var. = 0.037; N = 286,330.

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Appendix: What Is a "Bad" ED Doctor in the Literature?

Same patients, more resources, same health

- quasi-random assignment ⇒ gaps in tests, LOS, admit more
- flat of the curve: extra intensity, no better outcomes (Doyle et al. 2010; Silver 2021)

Low effort ⇒ more tests and more admissions

- cognitive load: tired doctors test more and admit more (Handel et al. 2026)
- end of shift: more admissions and more tests (Chan 2018)

Our contractors: more tests, longer stays, more admissions (+1–2pp), more handovers, same health